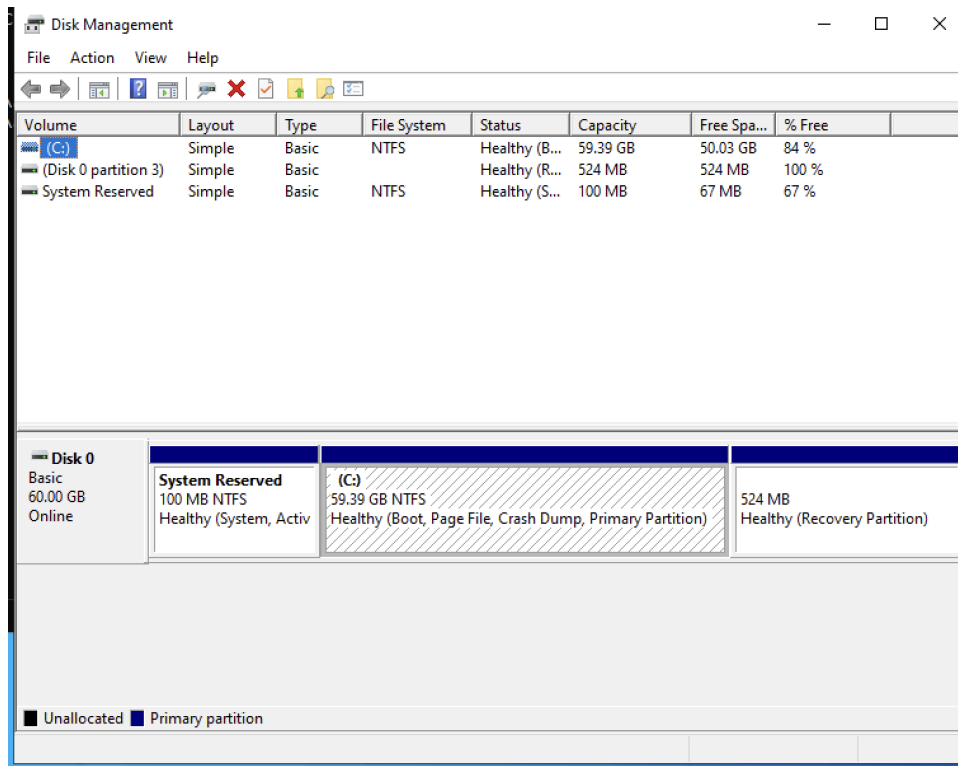


Omoze Idehenre
6/28/2026
PKI Week 13
Assets

Lab 1

My Week 13 VM contains only a single 60 GB virtual disk with no secondary disk or backup target. Since wbadmin requires a different target volume for System State backups, is this limitation expected, or should the lab VM include a second virtual disk?



```
Administrator: Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator.CORP> diskmgmt.msc
PS C:\Users\Administrator.CORP> certutil -CRL
CertUtil: -CRL command completed successfully.
PS C:\Users\Administrator.CORP>
```

Lab 2

```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Windows\system32> Get-ChildItem C:\CABackup -Recurse | Select FullName, Length, LastWriteTime

FullName                               Length  LastWriteTime
-----
C:\CABackup\DataBase                  4631    6/27/2026 4:33:44 AM
C:\CABackup\CVI Issuing CA 1.pl2      398     6/27/2026 4:33:44 AM
C:\CABackup\DataBase\certbkxp.dat     1052672 6/27/2026 4:33:44 AM
C:\CABackup\DataBase\CVI Issuing CA 1.edb 1048576 6/27/2026 4:33:44 AM
C:\CABackup\DataBase\edb00002.log      1048576 6/27/2026 4:33:44 AM
C:\CABackup\DataBase\edb00003.log      1048576 6/27/2026 4:33:44 AM

PS C:\Windows\system32>
```

```
Administrator: Windows PowerShell

PS C:\Windows\system32> certutil -ping
Connecting to PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1 ...
Server "CVI Issuing CA 1" ICertRequest2 interface is alive (62ms)
CertUtil: -ping command completed successfully.
PS C:\Windows\system32> certutil -CRL
CertUtil: -CRL command completed successfully.
PS C:\Windows\system32> certutil -store My
My "Personal"

===== Certificate 0 =====
Serial Number: 4400000003bf8819a444703a75000000000003
Issuer: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
NotBefore: 5/17/2026 7:23 AM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=cviissuingcal.corp.cvilab.local
Non-root Certificate
Template: CVI-WebServer
Cert Hash(sha1): 697926d7522ad1d3b1e0e5b271579e37fa45f4
Key Container = 0ff881e482ec5b0a91e962d3f817606e_f0a99c17-76d3-498a-97de-2992c06105fd
Simple container name: te-CVI-WebServer-6de1a47b-c00a-4c85-9a28-6539c8cccb8a
Provider = Microsoft RSA SChannel Cryptographic Provider
Private key is NOT exportable
Encryption test passed

===== Certificate 1 =====
Serial Number: 5800000002f7714edc7f317c460000000000002
Issuer: CN=CVI Root CA, DC=corp, DC=cvilab, DC=local
NotBefore: 4/25/2026 7:26 PM
NotAfter: 4/25/2027 7:36 PM
Subject: CN=CVI Issuing CA 1, DC=corp, DC=cvilab, DC=local
CA Version: V0.0
Certificate Template Name (Certificate Type): SubCA
Non-root Certificate
Template: SubCA, Subordinate Certification Authority
Cert Hash(sha1): 5137a597de2c3085ec3816c7f11edc18cfdabf8
Key Container = CVI Issuing CA 1
Unique container name: b52f658bb3f263e6f529f3a0187c63bc_f0a99c17-76d3-498a-97de-2992c06105fd
Provider = Microsoft Software Key Storage Provider
Signature test passed
CertUtil: -store command completed successfully.
PS C:\Windows\system32>
```

```
onName,NotAfter"
Schema:
-----
Column Name      Localized Name      Type      MaxLength
-----
RequestID        Issued Request ID   Long      4 -- Indexed
SerialNumber     Serial Number       String    128 -- Indexed
CommonName       Issued Common Name  String    8192 -- Indexed
NotAfter         Certificate Expiration Date  Date      8 -- Indexed

Row 1:
Issued Request ID: 0x3
Serial Number: "4400000003bf8819a444703a75000000000003"
Issued Common Name: "cviissuingcal.corp.cvilab.local"
Certificate Expiration Date: 4/25/2027 7:36 PM

Row 2:
Issued Request ID: 0x6
Serial Number: "4400000006165d0a6376a728ae000000000006"
Issued Common Name: "Users
Administrator"
Certificate Expiration Date: 4/25/2027 7:36 PM

Row 3:
Issued Request ID: 0x7
Serial Number: "440000000726eefaab089774b4000000000007"
Issued Common Name: "Users
Administrator"
Certificate Expiration Date: 4/25/2027 7:36 PM

Maximum Row Index: 3

3 Rows
12 Row Properties, Total Size = 402, Max Size = 76, Ave Size = 33
0 Request Attributes, Total Size = 0, Max Size = 0, Ave Size = 0
0 Certificate Extensions, Total Size = 0, Max Size = 0, Ave Size = 0
12 Total Fields, Total Size = 402, Max Size = 76, Ave Size = 33
CertUtil: -view command completed successfully.
PS C:\Windows\system32>
```

```
Windows PowerShell
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```

```
PS C:\Users\Administrator.CORP> Stop-Service CertSvc
PS C:\Users\Administrator.CORP> Remove-Item "C:\Windows\System32\CertLog\*.edb" -Force
PS C:\Users\Administrator.CORP> Remove-Item "C:\Windows\System32\CertLog\*.log" -Force
PS C:\Users\Administrator.CORP> dir C:\Windows\System32\CertLog
```

Directory: C:\Windows\System32\CertLog

Mode	LastWriteTime	Length	Name
-a----	6/27/2026 10:53 PM	16384	CVI Issuing CA 1.jfm
-a----	6/27/2026 10:53 PM	8192	edb.chk
-a----	4/25/2026 7:45 PM	1048576	edbres00001.jrs
-a----	4/25/2026 7:45 PM	1048576	edbres00002.jrs

```
PS C:\Users\Administrator.CORP>
```

Administrator: Windows PowerShell

Directory: C:\Windows\System32\CertLog

Mode	LastWriteTime	Length	Name
-a----	6/27/2026 10:53 PM	16384	CVI Issuing CA 1.jfm
-a----	6/27/2026 10:53 PM	8192	edb.chk
-a----	4/25/2026 7:45 PM	1048576	edbres00001.jrs
-a----	4/25/2026 7:45 PM	1048576	edbres00002.jrs

```
PS C:\Users\Administrator.CORP> Start-Service CertSvc
PS C:\Users\Administrator.CORP> Stop-Service CertSvc
PS C:\Users\Administrator.CORP> Remove-Item "C:\Windows\System32\CertLog\*" -Force
PS C:\Users\Administrator.CORP> dir C:\Windows\System32\CertLog
PS C:\Users\Administrator.CORP> Start-Service CertSvc
PS C:\Users\Administrator.CORP> Get-Service CertSvc
```

Status	Name	DisplayName
Stopped	CertSvc	Active Directory Certificate Services

```
PS C:\Users\Administrator.CORP> certutil -ping
Connecting to PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1 ...
Server could not be reached: The RPC server is unavailable. 0x800706ba (WIN32: 1722 RPC_S_SERVER_UNAVAI
```

```
CertUtil: -ping command FAILED: 0x800706ba (WIN32: 1722 RPC_S_SERVER_UNAVAILABLE)
CertUtil: The RPC server is unavailable.
```

```
PS C:\Users\Administrator.CORP>
```

Administrator: Windows PowerShell

```
-a----          6/27/2026    4:33 AM          1052672 CVI Issuing CA 1.edb
-a----          6/27/2026    4:33 AM          1048576 edb00002.log
-a----          6/27/2026    4:33 AM          1048576 edb00003.log
```

PS C:\Users\Administrator.CORP> dir C:\CABackup

Directory: C:\CABackup

Mode	LastWriteTime	Length	Name
d----	6/27/2026 4:33 AM		DataBase
-a----	6/27/2026 4:33 AM	4631	CVI Issuing CA 1.p12

```
PS C:\Users\Administrator.CORP> certutil -restoredb C:\CABackup
Restoring database for PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1.
Restoring Database files: 0%CertUtil: -restoreDB command FAILED: 0x80070091 (WIN32/HTTP: 145 ERROR_DIR_
CertUtil: The directory is not empty.
PS C:\Users\Administrator.CORP> Stop-Service CertSvc -Force
PS C:\Users\Administrator.CORP> Remove-Item "C:\Windows\System32\CertLog\*" -Force
PS C:\Users\Administrator.CORP> dir C:\Windows\System32\CertLog
PS C:\Users\Administrator.CORP> certutil -restoredb C:\CABackup\Database
Restoring database for PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1.
Not a valid backup directory: C:\CABackup\Database.
CertUtil: -restoreDB command FAILED: 0x80070002 (WIN32: 2 ERROR_FILE_NOT_FOUND)
CertUtil: The system cannot find the file specified.
PS C:\Users\Administrator.CORP>
```

Administrator: Windows PowerShell

Mode	LastWriteTime	Length	Name
d----	6/27/2026 4:33 AM		DataBase
-a----	6/27/2026 4:33 AM	4631	CVI Issuing CA 1.p12

```
PS C:\Users\Administrator.CORP> certutil -restoredb C:\CABackup
Restoring database for PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1.
Restoring Database files: 0%CertUtil: -restoreDB command FAILED: 0x80070091 (WIN32/HTTP: 145 ERROR_DIR_
CertUtil: The directory is not empty.
PS C:\Users\Administrator.CORP> Stop-Service CertSvc -Force
PS C:\Users\Administrator.CORP> Remove-Item "C:\Windows\System32\CertLog\*" -Force
PS C:\Users\Administrator.CORP> dir C:\Windows\System32\CertLog
PS C:\Users\Administrator.CORP> certutil -restoredb C:\CABackup\Database
Restoring database for PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1.
Not a valid backup directory: C:\CABackup\Database.
CertUtil: -restoreDB command FAILED: 0x80070002 (WIN32: 2 ERROR_FILE_NOT_FOUND)
CertUtil: The system cannot find the file specified.
PS C:\Users\Administrator.CORP> certutil -restorekey C:\CABackup
Enter PFX password:
CertUtil: -restoreKey command FAILED: 0x80070056 (WIN32: 86 ERROR_INVALID_PASSWORD)
CertUtil: The specified network password is not correct.
PS C:\Users\Administrator.CORP> certutil -f -restoredb C:\CABackup
Restoring database for PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1.
Restoring Database files: 100%
Restoring Log files: 100%
Full database restore for PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1.
Stop and Start Active Directory Certificate Services to complete database restore from C:\CABackup.
CertUtil: -restoreDB command completed successfully.
The CertSvc service may need to be restarted for changes to take effect.
PS C:\Users\Administrator.CORP>
```

```
Administrator: Windows PowerShell
Not a valid backup directory: C:\CABackup\Database.
CertUtil: -restoreDB command FAILED: 0x80070002 (WIN32: 2 ERROR_FILE_NOT_FOUND)
CertUtil: The system cannot find the file specified.
PS C:\Users\Administrator.CORP> certutil -restorekey C:\CABackup
Enter PFX password:
CertUtil: -restoreKey command FAILED: 0x80070056 (WIN32: 86 ERROR_INVALID_PASSWORD)
CertUtil: The specified network password is not correct.
PS C:\Users\Administrator.CORP> certutil -f -restoredb C:\CABackup
Restoring database for PKI-SRV01.corp.cvilib.local\CVI Issuing CA 1.
Restoring Database files: 100%
Restoring Log files: 100%
Full database restore for PKI-SRV01.corp.cvilib.local\CVI Issuing CA 1.
Stop and Start Active Directory Certificate Services to complete database restore from C:\CABackup.
CertUtil: -restoreDB command completed successfully.
The CertSvc service may need to be restarted for changes to take effect.
PS C:\Users\Administrator.CORP> Stop-Service CertSvc -Force
PS C:\Users\Administrator.CORP> Start-Service CertSvc
PS C:\Users\Administrator.CORP> Get-Service CertSvc

Status      Name              DisplayName
-----
Running     CertSvc           Active Directory Certificate Services

PS C:\Users\Administrator.CORP> certutil -ping
Connecting to PKI-SRV01.corp.cvilib.local\CVI Issuing CA 1 ...
Server "CVI Issuing CA 1" ICertRequest2 interface is alive (94ms)
CertUtil: -ping command completed successfully.
PS C:\Users\Administrator.CORP> certutil -view
Schema:
```

Lab 3

```
Administrator: Windows PowerShell
Windows PowerShell
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PS C:\Users\Administrator.CORP> Get-ChildItem C:\CABackup -Recurse | Select-Object FullName, Length, LastWriteTime

FullName                                     Length  LastWriteTime
-----
C:\CABackup\DataBase                        4631    6/27/2026 4:33:44 AM
C:\CABackup\CVI Issuing CA 1.p12           398     6/27/2026 4:33:44 AM
C:\CABackup\DataBase\certbkxp.dat          1052672 6/27/2026 4:33:44 AM
C:\CABackup\DataBase\CVI Issuing CA 1.edb  1048576 6/27/2026 4:33:44 AM
C:\CABackup\DataBase\edb00002.log          1048576 6/27/2026 4:33:44 AM
C:\CABackup\DataBase\edb00003.log          1048576 6/27/2026 4:33:44 AM

PS C:\Users\Administrator.CORP> Stop-Service CertSvc
PS C:\Users\Administrator.CORP> Get-Service CertSvc

Status      Name              DisplayName
-----
Stopped     CertSvc           Active Directory Certificate Services

PS C:\Users\Administrator.CORP>
```

```

FullName                               Length  LastWriteTime
-----
C:\CABackup\DataBase                  6/27/2026 4:33:44 AM
C:\CABackup\CVI Issuing CA 1.p12      4631    6/27/2026 4:33:44 AM
C:\CABackup\DataBase\certbkxp.dat      398     6/27/2026 4:33:44 AM
C:\CABackup\DataBase\CVI Issuing CA 1.edb 1052672 6/27/2026 4:33:44 AM
C:\CABackup\DataBase\edb00002.log      1048576 6/27/2026 4:33:44 AM
C:\CABackup\DataBase\edb00003.log      1048576 6/27/2026 4:33:44 AM

```

```

PS C:\Users\Administrator.CORP> Stop-Service CertSvc
PS C:\Users\Administrator.CORP> Get-Service CertSvc

```

```

Status  Name          DisplayName
-----
Stopped CertSvc      Active Directory Certificate Services

```

```

PS C:\Users\Administrator.CORP> Remove-Item "C:\Windows\System32\CertLog\*" -Force -Recurse
PS C:\Users\Administrator.CORP> dir C:\Windows\System32\CertLog
PS C:\Users\Administrator.CORP> Start-Service CertSvc
PS C:\Users\Administrator.CORP> Get-Service CertSvc

```

```

Status  Name          DisplayName
-----
Stopped CertSvc      Active Directory Certificate Services

```

```

PS C:\Users\Administrator.CORP>

```

```

Status  Name          DisplayName
-----
Running CertSvc      Active Directory Certificate Services
Connecting to PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1 ...
Server could not be reached: The RPC server is unavailable. 0x800706ba (WIN32: 1722 RPC_S_SERVER_UNAVAILA
CertUtil: -ping command FAILED: 0x800706ba (WIN32: 1722 RPC_S_SERVER_UNAVAILABLE)
CertUtil: The RPC server is unavailable.

```

```

PS C:\Users\Administrator.CORP> Restart-Service CertSvc -Force
PS C:\Users\Administrator.CORP> Start-Sleep -Seconds 10
PS C:\Users\Administrator.CORP> certutil -ping
Connecting to PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1 ...
Server "CVI Issuing CA 1" ICertRequest2 interface is alive (63ms)
CertUtil: -ping command completed successfully.
PS C:\Users\Administrator.CORP>

```

```

Status  Name          DisplayName
-----
Running CertSvc      Active Directory Certificate Services
Connecting to PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1 ...
Server could not be reached: The RPC server is unavailable. 0x800706ba (WIN32: 1722 RPC_S_SERVER_UNAVAILA
CertUtil: -ping command FAILED: 0x800706ba (WIN32: 1722 RPC_S_SERVER_UNAVAILABLE)
CertUtil: The RPC server is unavailable.

```

```

PS C:\Users\Administrator.CORP> Restart-Service CertSvc -Force
PS C:\Users\Administrator.CORP> Start-Sleep -Seconds 10
PS C:\Users\Administrator.CORP> certutil -ping
Connecting to PKI-SRV01.corp.cvilab.local\CVI Issuing CA 1 ...
Server "CVI Issuing CA 1" ICertRequest2 interface is alive (63ms)
CertUtil: -ping command completed successfully.
PS C:\Users\Administrator.CORP>

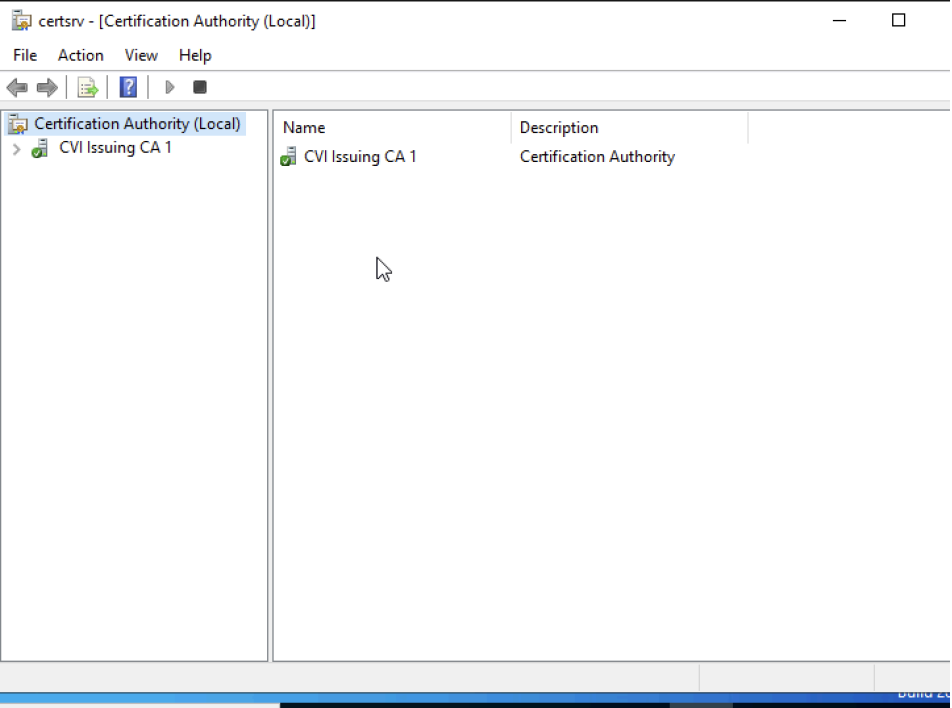
```

```
Windows PowerShell
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PS C:\Users\Administrator.CORP> Get-Service CertSvc
```

Status	Name
Running	CertSvc



certsrv - [Certification Authority (Local)]

File Action View Help

Name	Description
CVI Issuing CA 1	Certification Authority